

Global Semiconductors and Semiconductor Capital Equipment

Global Semis: July 2026 WSTS Tracker - Sales -9.5% MoM, a bit worse than typical (-8.5% MoM) but +131% YoY on continued memory strength



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On Friday September 4 the SIA published its monthly WSTS report, which tracks sales, units and ASPs across the semiconductor industry, and provides details by products, end markets, and geographies. The recent report includes data up to and including July 2026.

Total semiconductor sales grew 131% YoY in July after rising 144% YoY in June. YoY strength was again dominated by memory with sales growing 452% YoY, driving ~\$355B in incremental revenue YTD and with memory pricing alone driving almost 70% of the 111% total industry growth seen since the beginning of the year. Excluding memory, sales were still up strongly at 35% YoY.

Total semiconductor sales MoM were a touch below typical seasonality, down-9.5% versus the historical July average of -8.5% MoM. On a MoM basis as-reported, product group performance was generally better than seasonal patterns.

Sales were better than typical MoM for Discretes (-5.9% vs. typical -10.2%), Analog Std Linear (8.6% vs. typical 2.7%), Logic (2.2% vs. typical -0.9%), MCU (-1.5% vs. typical -9.1%), DSP (6.8% vs. typical -5.1%), DRAM (-17.5% vs. typical -25.7%), NAND (-14.7% vs. typical -28.8%). MPU (-8.2% vs. typical -5.7%) was below typical MoM patterns, while Optoelectronics (3.9% vs. typical 3.6%), Sensors & Actuators (-4.6% vs. typical -4.4%), Analog App Specific (0.0% vs. typical 0.0%) were in line with historical patterns.

By geography, total sales increased YoY in all regions, and MoM changes were mixed.

Total unit shipments were flattish at -0.3% MoM, with ASPs down -9.3% MoM. On a YoY basis, shipments increased by 16.3%, and industry ASPs rose 98.8% YoY on the back of memory price increases.

Application specific ICs sales were up 0.5% MoM, better than typical historical patterns (down -2.5% MoM). 3 end markets were better than typical MoM while 3 were worse than typical. Groups that performed better than historical MoM patterns were: Computer & Peripherals (2.8% vs. typical -3.0%), Automotive (-0.5% vs. typical -5.3%), Multipurpose and Others (-3.6% vs. typical -5.6%). Meanwhile, Consumer (-8.2% vs. typical -6.0%), Wireless Communication (-6.1% vs. typical -0.1%), Wired Communication (1.2% vs. typical 3.9%) were worse than typical MoM patterns.

BERNSTEIN TICKER TABLE

Ticker	Rating	4 Sep 2026		Price Target	TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
		Cur	Closing Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
AMD (Advanced Micro)	O	USD	477.57	650.00	197.1%	USD	4.17	7.55	15.56	114.4	63.3	30.7
ADI (Analog Devices)	O	USD	362.25	465.00	27.8%	USD	7.79	12.85	16.77	46.5	28.2	21.6
AVGO (Broadcom)	O	USD	357.90	575.00	(12.0)%	USD	6.82	11.64	18.63	52.5	30.7	19.2
INTC (Intel)	M	USD	95.80	110.00	272.3%	USD	0.43	1.55	1.81	224.7	61.9	52.8
NVDA (NVIDIA)	O	USD	230.36	400.00	19.1%	USD	4.77	9.20	15.80	48.3	25.0	14.6
NXPI (NXP Semiconductors)	M	USD	227.84	290.00	(18.4)%	USD	11.81	15.02	16.85	19.3	15.2	13.5
QCOM (Qualcomm)	M	USD	168.74	165.00	(13.3)%	USD	12.03	10.51	9.57	14.0	16.1	17.6
TXN (Texas Instruments)	M	USD	258.44	290.00	18.7%	USD	5.45	8.33	9.16	47.4	31.0	28.2
PDFS (PDF Solutions)	O	USD	45.65	65.00	110.0%	USD	0.94	1.35	1.75	48.5	33.9	26.1
KLAC	O	USD	185.60	250.00	86.2%	USD	3.76	5.47	7.05	49.4	33.9	26.3
LRCX (Lam Research)	O	USD	307.65	385.00	180.0%	USD	5.82	9.22	10.97	52.9	33.4	28.1
AMAT (Applied Materials)	O	USD	454.71	700.00	160.6%	USD	9.42	12.76	18.30	48.3	35.6	24.8
TSM (TSMC)	O	USD	428.91	554.00	57.4%	USD	10.28	17.18	22.48	13.2	9.5	7.1
2330.TT (TSMC)	O	TWD	2,460.00	3,300.00	76.6%	TWD	66.25	110.70	144.85	11.8	8.4	6.3
2303.TT (UMC)	U	TWD	143.00	88.00	211.9%	TWD	3.34	4.58	5.56	4.7	4.3	4.0
5347.TT (Vanguard)	M	TWD	156.00	146.00	37.9%	TWD	4.30	5.70	6.59	4.6	4.3	4.0
2454.TT (MediaTek)	O	TWD	4,760.00	4,380.00	201.0%	TWD	66.17	69.52	170.50	71.9	68.5	27.9
3034.TT (Novatek)	M	TWD	540.00	480.00	(8.3)%	TWD	26.87	30.43	32.54	20.1	17.7	16.6
005930.KS (SEC- Samsung)	O	KRW	269,500	440,000	252.6%	KRW	6,611.53	48,393	77,273	40.8	5.6	3.5
005935.KS (SEC-Pref - Samsung)	O	KRW	199,600	374,000	215.2%	KRW	6,611.53	48,393	77,273	30.2	4.1	2.6
SMSN.LI (Samsung)	O	USD	5,045.00	7,350.00	272.3%	USD	116.15	812.39	1,290.80	43.4	6.2	3.9
000660.KS (SK hynix)	O	KRW	1,787,000	3,300,000	513.2%	KRW	60,341	395,677	568,862	29.6	4.5	3.1
MU (Micron)	O	USD	1,016.59	1,300.00	655.0%	USD	8.29	67.39	158.99	122.7	15.1	6.4
285A.JP (KIOXIA)	U	JPY	59,530	40,000	1758.0%	JPY	1,014.00	10,013	9,656.84	58.7	5.9	6.2
688012.CH (AMEC)	O	CNY	344.00	660.00	109.1%	CNY	2.28	5.30	7.26	150.8	65.0	47.4
688041.CH (Hygon)	O	CNY	242.71	450.00	3.4%	CNY	1.10	2.04	3.59	220.6	118.8	67.5
002371.CH (NAURA)	O	CNY	660.65	1,380.00	46.5%	CNY	5.66	10.11	19.92	116.7	65.4	33.2
688072.CH (Piotech)	O	CNY	651.92	1,200.00	237.1%	CNY	3.32	10.18	16.61	196.4	64.0	39.2
6415.TT (Silergy)	O	TWD	418.00	650.00	8.4%	TWD	6.40	9.54	16.69	65.3	43.8	25.0
ASML.NA (ASML)	O	EUR	1,500.40	2,500.00	109.5%	EUR	24.72	38.91	53.56	60.7	38.6	28.0
ASML (ASML)	O	USD	1,714.88	2,859.00	100.5%	USD	27.95	44.01	60.58	52.8	33.5	24.3
6857.JP (Advantest)	O	JPY	34,480	45,800	154.6%	JPY	513.30	948.96	1,088.87	67.2	36.3	31.7
6146.JP (DISCO)	O	JPY	57,150	99,000	17.3%	JPY	1,245.79	1,830.28	2,336.94	45.9	31.2	24.5
7741.JP (Hoya)	O	JPY	24,145	35,500	(17.8)%	JPY	744.42	829.63	977.78	32.4	29.1	24.7
4062.JP (Ibiden)	O	JPY	22,200	29,000	477.5%	JPY	214.91	329.25	511.84	103.3	67.4	43.4
6525.JP (Kokusai)	O	JPY	8,929.00	12,420	191.1%	JPY	128.63	262.29	373.69	69.4	34.0	23.9
6920.JP (Lasertec)	O	JPY	35,590	54,000	88.3%	JPY	863.65	1,185.31	1,348.53	41.2	30.0	26.4
6723.JP (Renesas)	O	JPY	3,403.00	6,300.00	65.8%	JPY	181.61	261.36	302.64	18.7	13.0	11.2
3436.JP (SUMCO)	M	JPY	3,315.00	3,290.00	123.0%	JPY	(33.61)	(40.64)	70.73	(98.6)	(81.6)	46.9
7735.JP (Screen)	M	JPY	13,515	15,000	106.9%	JPY	486.61	574.49	759.55	27.8	23.5	17.8
8035.JP (Tokyo Electron)	O	JPY	55,840	79,300	136.9%	JPY	1,250.88	1,687.17	2,635.81	44.6	33.1	21.2
SOI.FP (Soitec)	O	EUR	143.35	200.00	316.4%	EUR	(6.17)	(0.32)	3.59	(23.2)	(450.6)	39.9
XFAB.FP (X-Fab)	O	EUR	6.35	12.80	(20.6)%	USD	0.23	0.07	0.51	19.4	50.7	15.4
SNDK (SanDisk)	O	USD	1,740.00	3,000.00	2419.5%	USD	70.08	238.36	272.09	24.8	7.3	6.4
SPX			7,718.60									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

AMD, ADI, AVGO, INTC, NVDA, NXPI, QCOM, PDFS, KLAC, LRCX, AMAT, MU, 6723.JP, SOI.FP, SNDK estimate is Adjusted EPS; AMD, ADI, AVGO, INTC, NVDA, NXPI, QCOM, PDFS, KLAC, LRCX, AMAT, MU, 6723.JP, SOI.FP, SNDK valuation is Adjusted P/E (x); TSM, 2330.TT, 2303.TT, 5347.TT valuation is P/B (x); XFAB.FP valuation is EV/EBIT (x); NVDA, KLAC, LRCX, 6920.JP, SOI.FP, SNDK base year is 2026;

Source: Bloomberg, Bernstein estimates and analysis.

Ticker	Rating	Cur	4 Sep 2026		TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
			Closing Price	Price Target		Cur	2025A	2026E	2027E	2025A	2026E	2027E
ASIA			2,003.84									
EM			1,863.45									
EDME			1,611.88									
JPL			2,689.74									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

AMD, ADI, AVGO, INTC, NVDA, NXPI, QCOM, PDFS, KLAC, LRCX, AMAT, MU, 6723.JP, SOI.FP, SNDK estimate is Adjusted EPS; AMD, ADI, AVGO, INTC, NVDA, NXPI, QCOM, PDFS, KLAC, LRCX, AMAT, MU, 6723.JP, SOI.FP, SNDK valuation is Adjusted P/E (x); TSM, 2330.TT, 2303.TT, 5347.TT valuation is P/B (x); XFAB.FP valuation is EV/EBIT (x); NVDA, KLAC, LRCX, 6920.JP, SOI.FP, SNDK base year is 2026;

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

ADI (Outperform, \$465.00): ADI is executing well with both cyclical and idiosyncratic drivers, and valuation now suggesting better risk-reward.

AMD (Outperform, \$650.00): Expectations remain high, but exposure to AI demand driving both a CPU and GPU story can provide substantial growth.

AVGO (Outperform, \$755.00): A strong 2026 AI trajectory seems set to markedly accelerate into 2027 and 2028.

INTC (Market-Perform, \$110.00): Server strength is helping the company get back on their feet, and narrative/headlines may fuel the vibe for now

NVDA (Outperform, \$400.00): The datacenter opportunity is enormous..

NXPI (Market-Perform, \$290.00): The pace and makeup of recovery remains open for debate, though company execution is solid and valuation is increasingly inexpensive.

QCOM (Market-Perform, \$165.00): The diversification story is real but a number of near-term headwinds (smartphone memory dynamics, margins, AAPL etc) need to be navigated first.

TXN (Market-Perform, \$290.00): TXN shares feel fully valued in the current environment.

PDFS (Outperform, \$65.00) A transition to analytics provides for growth and margin expansion amid numerous structural drivers, with eProbe tools representing a new meaningful business opportunity.

AMAT (Outperform, \$700.00): Exposure to key inflections is strong & valuation vs peers is attractive

LRCX (Outperform, \$385.00): The company is benefiting from key inflections (GAA, packaging, HBM, NAND upgrades) and CY26/CY27 commentary seems supportive.

KLAC (Outperform, \$250.00): Amid positive WFE trends KLAC possess structural growth drivers, a strong and durable competitive position, lower China replacement risk, and disciplined capital allocation, warranting premium valuation.

TSMC: We rate TSMC Outperform with PT=NT\$3,300.00.

UMC: We rate UMC Underperform with PT=NT\$88.00.

Vanguard: We rate Vanguard Market-Perform with PT=NT\$146.00.

Novatek: We rate Novatek Market-Perform with PT=NT\$480.00.

MediaTek: We rate MediaTek Outperform with PT=NT\$4,380.00.

Samsung Electronics: We rate Samsung Electronics Outperform with target price of KRW 440,000.

SK hynix: We rate SK hynix Outperform with target price to KRW 3,300,000.

Micron: We rate Micron Outperform with target price of US\$1,300.00.

KIOXIA: We rate KIOXIA Underperform with target price of JPY40,000.

SOI.FP: We rate Soitec Outperform with target price of €200.00.

XFAB.FP: We rate X-FAB Outperform with target price of €12.80

NAURA: We rate Naura Outperform with PT of CNY 1,380.00.

AMEC: We rate AMEC Outperform with PT of CNY 660.00.

Piotech: We rate Piotech Outperform with PT of CNY 1,200.00.

Hygon: We rate Hygon Outperform with PT of CNY 450.00.

Silergy: We rate Silergy Outperform with PT of TWD 650.00.

DISCO: We rate DISCO Outperform with PT of JPY 99,000.

Tokyo Electron: We rate Tokyo Electron Outperform with PT of JPY 79,300.

ASML: We rate ASML Outperform with PT of EUR 2,500.00.

Advantest: We rate Advantest Outperform with PT of JPY 45,800.

Screen: We rate Screen Market-Perform with PT of JPY 15,000.

Kokusai: We rate Kokusai Outperform with PT of JPY 12,420.

Lasertec: We rate Lasertec Outperform with PT of JPY 54,000.

SUMCO: We rate SUMCO Market-Perform with PT of JPY 3,290.00.

Hoya: We rate Hoya Outperform with PT of JPY 35,500.

Ibiden: We rate Ibiden Outperform with PT of JPY 29,000.

Renesas: We rate Renesas Outperform with PT of JPY 6,300.00.

SanDisk: We rate Sandisk Outperform with PT of US\$ 3,000.

DETAILS

On Friday September 4th the SIA published its monthly WSTS report, which tracks sales, units and ASPs across the semiconductor industry, and provides details by products, end markets, and geographies. The recent report includes data up to and including July 2026.

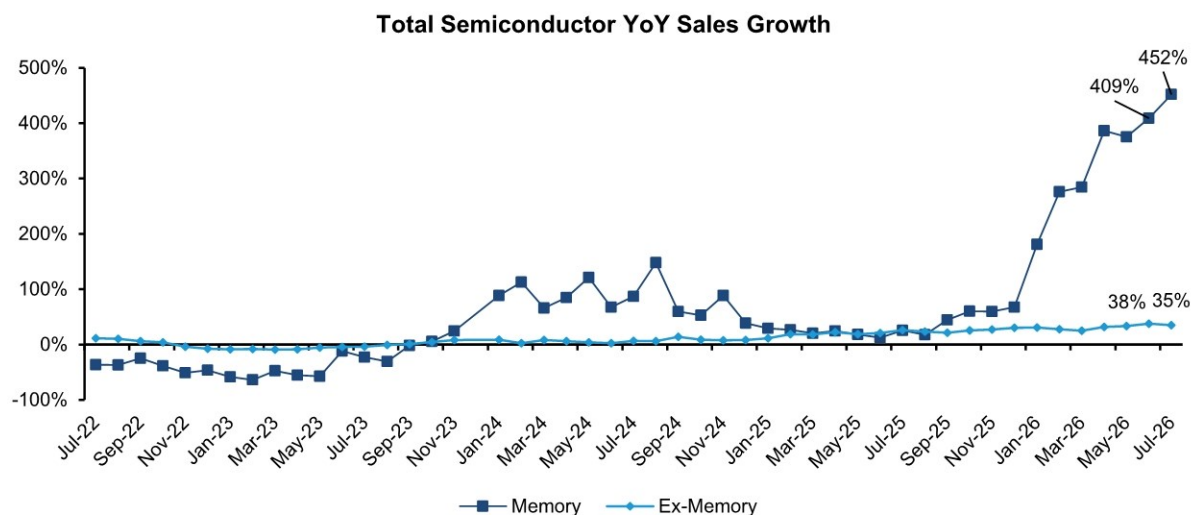
Total semiconductor sales grew 131% YoY in July after rising 144% YoY in June (**Exhibit 1**). YoY strength was again dominated by memory with sales growing 452% YoY, driving ~\$355B in incremental revenue YTD and with memory pricing alone driving almost 70% of the 111% total industry growth seen since the beginning of the year (**Exhibit 3, Exhibit 4**). Excluding memory, sales were still up strongly at 35% YoY (**Exhibit 2**). On a 3-month rolling basis sales were up 30.4%, compared to typical historical patterns of up MSD (5.5%) (**Exhibit 5**).

EXHIBIT 1: **Historical total semiconductor YoY sales growth**

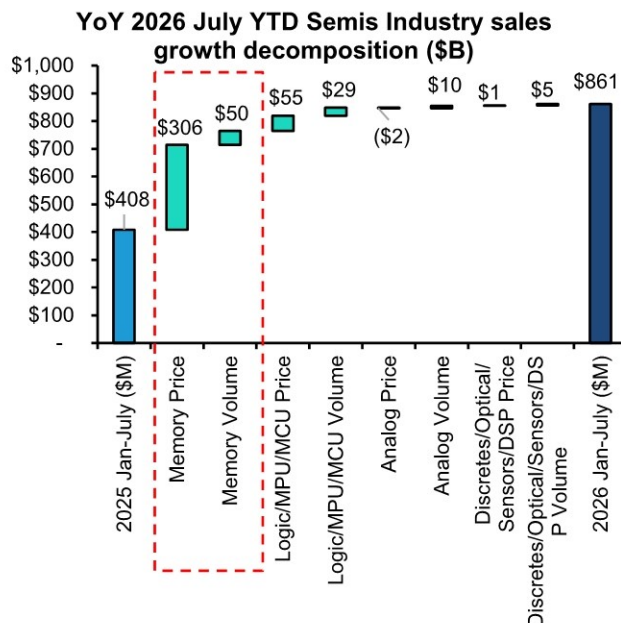


Source: WSTS, Bernstein Analysis

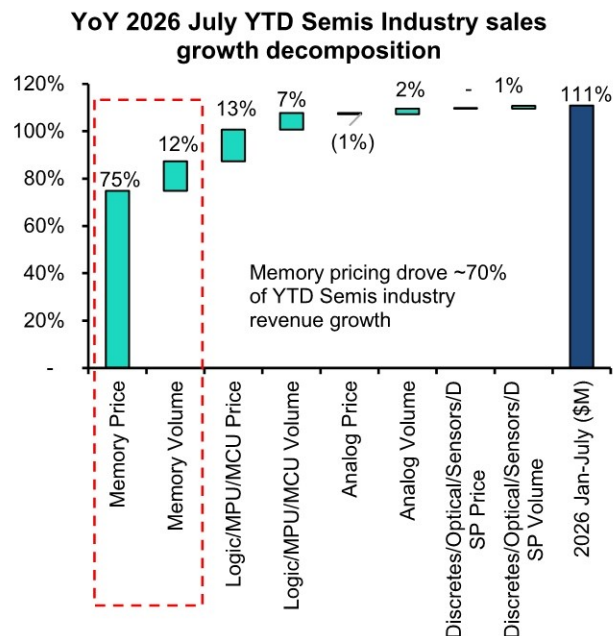
EXHIBIT 2: **Historical memory vs. non-memory YoY sales growth**



Source: WSTS, Bernstein Analysis

EXHIBIT 3: Memory drove ~\$355B in incremental Semis sales YTD through July...

Price x volume mix effects attributed to price.
Source: WSTS, Bernstein analysis

EXHIBIT 4: ...with Memory pricing alone driving almost 70% of the 111% total YTD Semis revenue growth

Price x volume mix effects attributed to price.
Source: WSTS, Bernstein analysis

EXHIBIT 5: Semiconductor industry % change in sales

	vs 1 month ago		vs 3 months ago		3 months rolling		vs 1 year ago
	This Month	Historical Average (20-25)	This Month	Historical Average (20-25)	This Month	Historical Average (20-25)	This month
Discretes	-5.9%	-10.2%	13.4%	7.9%	8.9%	4.3%	22.2%
Optoelectronics	3.9%	3.6%	3.1%	4.7%	2.6%	-2.1%	8.4%
Sensors & Actuators	-4.6%	-4.4%	6.9%	5.9%	5.1%	3.1%	1.3%
Analog Std Linear	8.6%	2.7%	18.6%	7.6%	12.3%	4.6%	34.4%
Analog App Specific	0.0%	0.0%	7.8%	10.0%	3.5%	2.9%	5.8%
Logic	2.2%	-0.9%	14.0%	7.5%	12.9%	6.0%	49.7%
MPU	-8.2%	-5.7%	18.8%	4.6%	20.7%	2.7%	36.3%
MCU	-1.5%	-9.1%	7.8%	5.3%	10.2%	3.2%	17.2%
DSP	6.8%	-5.1%	23.2%	6.6%	25.2%	-0.2%	3.2%
Memory	-16.3%	-26.1%	32.5%	3.6%	48.3%	12.5%	451.7%
DRAM	-17.5%	-25.7%	28.7%	7.8%	42.7%	12.7%	427.8%
NAND	-14.7%	-28.8%	39.9%	-0.9%	59.8%	13.6%	531.9%
TOTAL	-9.5%	-8.5%	23.0%	5.7%	30.4%	5.5%	131.4%

Source: WSTS, Bernstein Analysis

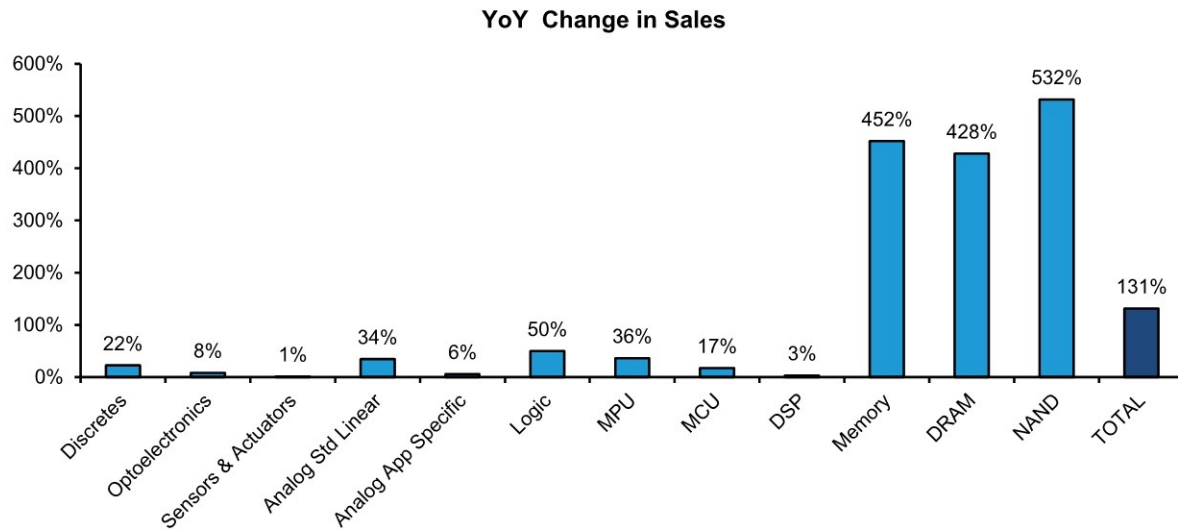
Total semiconductor sales MoM were below typical seasonality, down-9.5% versus the historical July average of -8.5% MoM. On a MoM basis as-reported, product group performance was generally better than seasonal patterns.

In July on a YoY basis and on a product group level, all product groups grew (**Exhibit 6**).

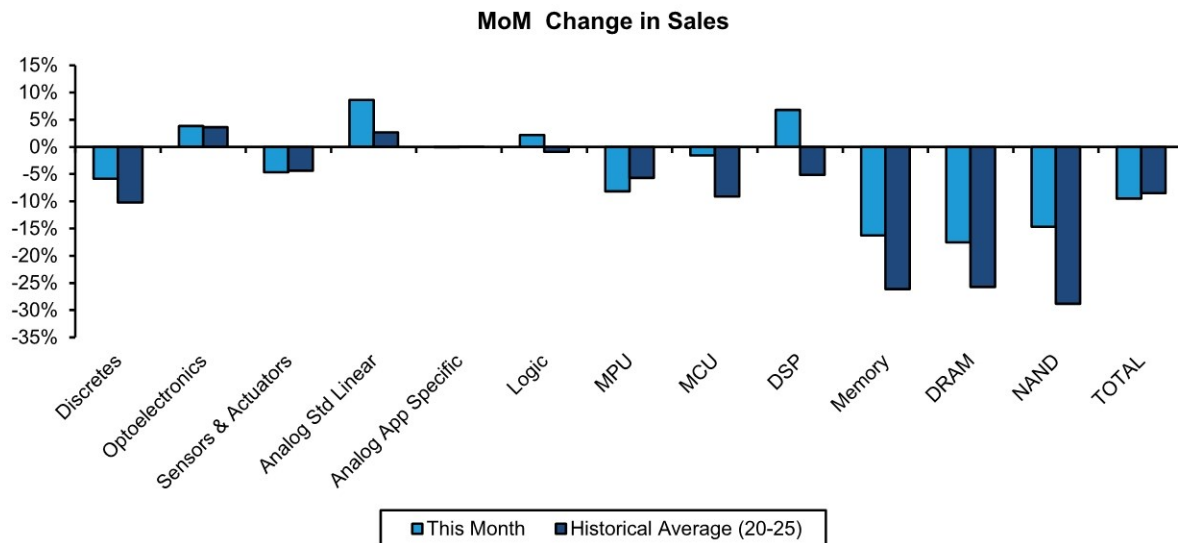
On a MoM basis, performance by product group was relatively positive with 7 groups better than typical, 3 groups in line, and 1 group worse than typical patterns. (**Exhibit 7**).

- Discretes (-5.9% vs. typical -10.2%), Analog Std Linear (8.6% vs. typical 2.7%), Logic (2.2% vs. typical -0.9%), MCU (-1.5% vs. typical -9.1%), DSP (6.8% vs. typical -5.1%), DRAM (-17.5% vs. typical -25.7%), NAND (-14.7% vs. typical -28.8%), were better than typical;

- MPU (-8.2% vs. typical -5.7%) was below typical MoM patterns, while Optoelectronics (3.9% vs. typical 3.6%), Sensors & Actuators (-4.6% vs. typical -4.4%), Analog App Specific (0.0% vs. typical 0.0%) were in line with historical patterns.

EXHIBIT 6: **Total semiconductor YoY change in sales**

Source: WSTS, Bernstein Analysis

EXHIBIT 7: **Total semiconductor MoM change in sales**

Source: WSTS, Bernstein Analysis

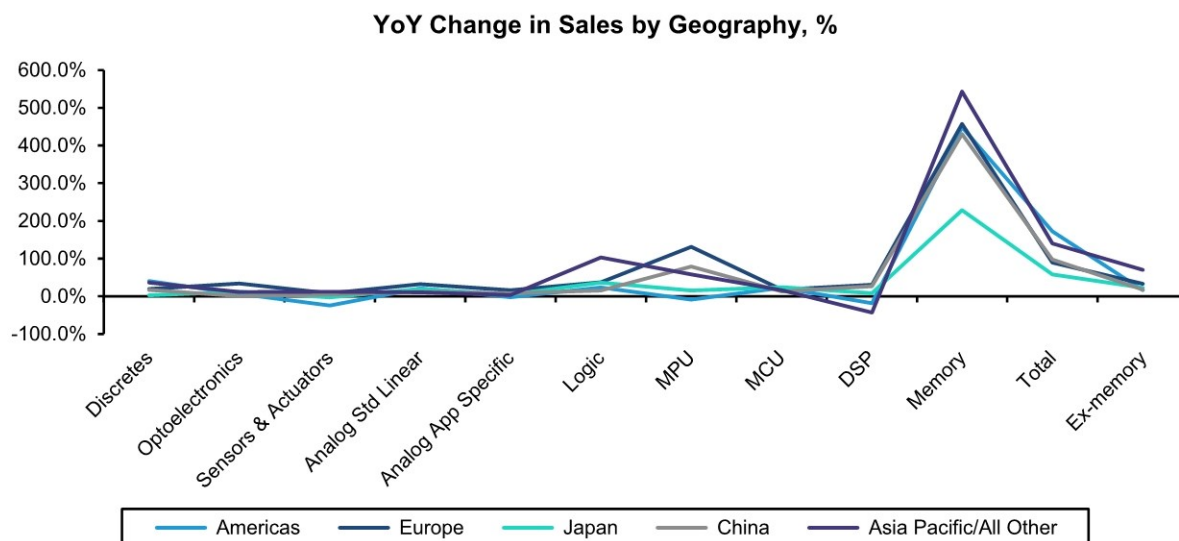
By geography, total sales increased YoY in all regions, and MoM changes were mixed.

On a YoY basis, total semiconductors sales were up 172.6% in the Americas, 89.8% in Europe, 58.4% in Japan, 97.5% in China, and 140.3% in Asia Pacific/All Other (**Exhibit 8**). Excluding memory, total sales for these regions were up 16.0%, 33.5%, 23.1%, 17.9%, 70.5%, respectively.

On a MoM basis, sales changes were -0.4% in the Americas, -4.4% in Europe, +1.6% in Japan, -28.0% in China, and -4.7% in Asia Pacific/All Other (**Exhibit 9**). Excluding memory, sales changes for these regions were +13.3%, -5.9%, +0.2%, -5.9%,

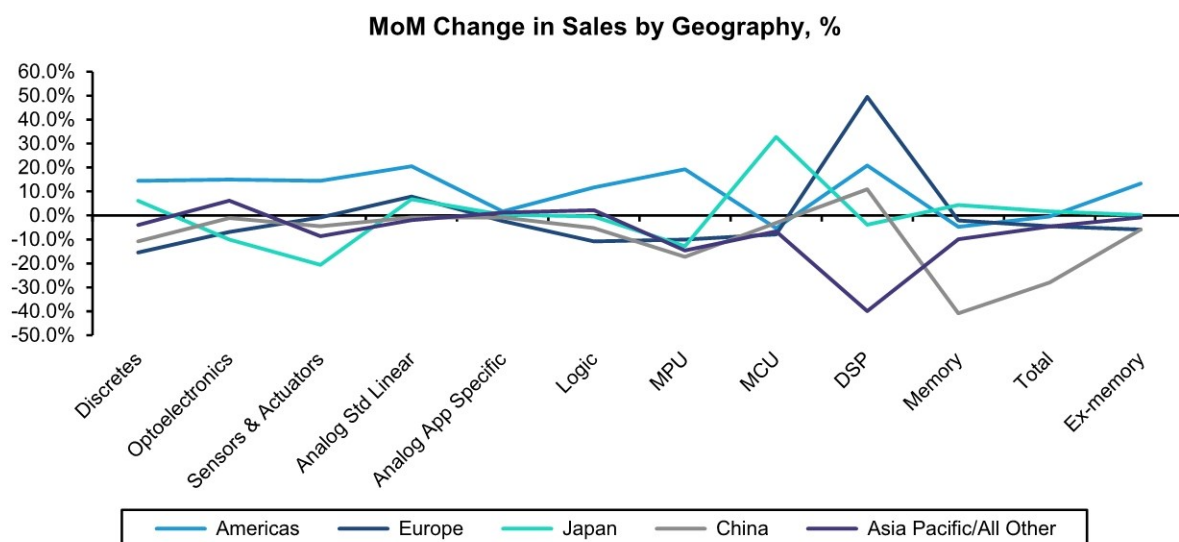
-0.9%, respectively.

EXHIBIT 8: **Total semiconductor YoY change in sales by geography**



Source: WSTS, Bernstein Analysis

EXHIBIT 9: **Total semiconductor MoM change in sales by geography**



Source: WSTS, Bernstein Analysis

Total unit shipments were flattish at -0.3% MoM, with ASPs down -9.3% MoM.

- 8 out of 11 products groups showed a decrease in unit shipments MoM while 3 groups had MoM unit increases.
- Units were up MoM for Discretes, Optoelectronics, DSP, (2.3%, 6.5%, 5.0% respectively).
- Units were down MoM for Sensors & Actuators, Analog Std Linear, Analog App Specific, Logic, MPU, MCU, DRAM[bits], NAND[bits], (-11.0%, -1.5%, -13.3%, -1.5%, -8.7%, -3.9%, -22.5%, -22.1% respectively)
- On a 3 months rolling basis overall shipments were up 10.2% and increased for all product groups.

- On a YoY basis, shipments increased by 16.3%. Units increased for all product groups except MCU (-0.7%). (**Exhibit 10**).

EXHIBIT 10: Change in semiconductor unit shipments

% Change in Units

	vs 1 month ago	vs 3 months ago	3 months rolling	vs 1 year ago
Discretes	2.3%	9.1%	8.8%	6.0%
Optoelectronics	6.5%	14.1%	10.3%	23.9%
Sensors & Actuators	-11.0%	8.1%	7.8%	6.2%
Analog Std Linear	-1.5%	14.0%	14.0%	30.3%
Analog App Specific	-13.3%	-1.5%	1.3%	11.4%
Logic	-1.5%	9.2%	12.4%	26.5%
MPU	-8.7%	18.8%	12.8%	16.6%
MCU	-3.9%	6.9%	8.1%	-0.7%
DSP	5.0%	10.8%	23.0%	15.7%
DRAM (Based on Bit Shipped)	-22.5%	2.3%	6.8%	47.5%
NAND (Based on Bit Shipped)	-22.1%	3.9%	7.5%	42.3%
TOTAL	-0.3%	10.3%	10.2%	16.3%

Source: WSTS, Bernstein Analysis

On a MoM basis, industry ASPs were down -9.3%. ASPs fell in 2 product groups, rose in 8 groups and were flattish in 1 MoM.

- ASPs increased MoM for Sensors & Actuators, Analog Std Linear, Analog App Specific, Logic, MCU, DSP, DRAM[bits], NAND[bits] (7.3%, 10.1%, 15.4%, 3.7%, 2.4%, 1.8%, 6.5%, 9.5% respectively). MPU ASP was roughly flat (0.5%). ASPs fell MoM for Discretes and Optoelectronics, (-7.0%, -2.4% respectively).
- Versus 3 months ago, 7 out of 11 product groups saw ASP increases, 2 flattish, and 2 seeing decreases, while on a YoY basis ASPs rose for 7 groups and fell for 4 groups. Total industry ASPs rose 98.8% YoY on the back of memory price increases. (**Exhibit 11**).

EXHIBIT 11: Change in semiconductor ASPs

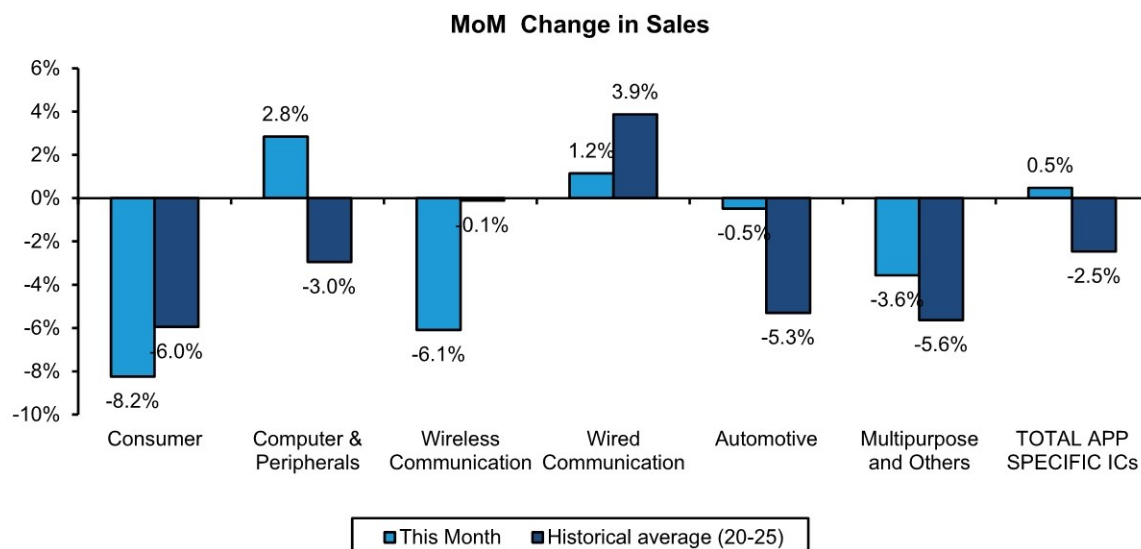
% Change in ASP

	vs 1 month ago	vs 3 months ago	vs 1 year ago
Discretes	-7.0%	4.8%	15.8%
Optoelectronics	-2.4%	-9.6%	-12.5%
Sensors & Actuators	7.3%	-1.0%	-4.7%
Analog Std Linear	10.1%	4.0%	3.2%
Analog App Specific	15.4%	9.4%	-5.0%
Logic	3.7%	4.5%	18.4%
MPU	0.5%	0.0%	16.9%
MCU	2.4%	0.9%	18.0%
DSP	1.8%	11.2%	-10.8%
DRAM (Based on \$/Bit)	6.5%	25.8%	257.9%
NAND (Based on \$/Bit)	9.5%	34.6%	344.1%
TOTAL	-9.3%	11.5%	98.8%

Source: WSTS, Bernstein Analysis

Application specific ICs sales were up 0.5% MoM, better than typical historical patterns (down -2.5% MoM). 3 end markets were better than typical MoM while 3 were worse than typical (Exhibit 12).

- 3 product groups were above typical MoM: Computer & Peripherals (2.8% vs. typical -3.0%), Automotive (-0.5% vs. typical -5.3%), Multipurpose and Others (-3.6% vs. typical -5.6%),
- 3 groups were worse than typical: Consumer (-8.2% vs. typical -6.0%), Wireless Communication (-6.1% vs. typical -0.1%), Wired Communication (1.2% vs. typical 3.9%)

EXHIBIT 12: **Change in sales by end markets (where WSTS data provides a split by application)**

Source: WSTS, Bernstein Analysis

By product category, we note the following:

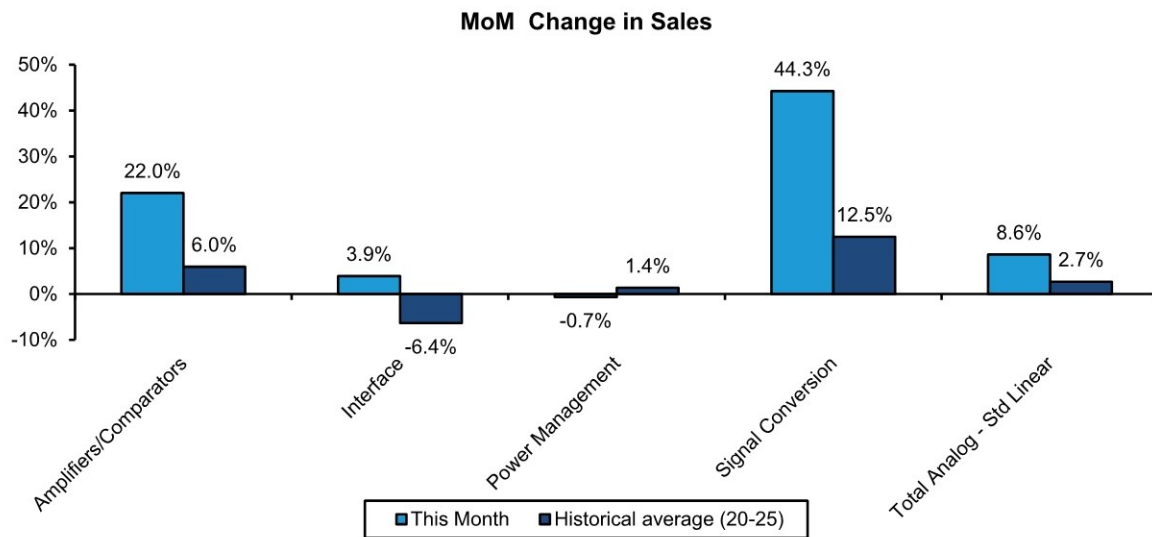
Analog Standard Linear: Total Analog Standard Linear sales were up MoM (8.6%), better than typical patterns (2.7%). Shipments were down(-1.5%) with ASPs up (10.1%). Product category changes MoM were as follows: Amplifiers/Comparators (22.0% vs. typical 6.0%), Interface (3.9% vs. typical -6.4%), Power Management (-0.7% MoM vs. typical 1.4% MoM), and Signal Conversion (44.3% vs. typical 12.5%) (**Exhibit 13**).

Analog Application Specific: Total analog application specific sales were flattish (0.0%) MoM, in-line with typical of 0.0% MoM. Units were down (-13.3%) MoM, with ASPs up (15.4%). End markets changes MoM were as follows: Consumer (-20.3% MoM vs. typical -2.6% MoM), Computer (-8.8% vs. typical -3.4%), Communications (5.7% MoM vs. typical 6.3% MoM), Automotive (0.7% MoM vs. typical -5.8% MoM), and Industrial & Others (-5.0% vs. typical of -3.5%)(**Exhibit 14**).

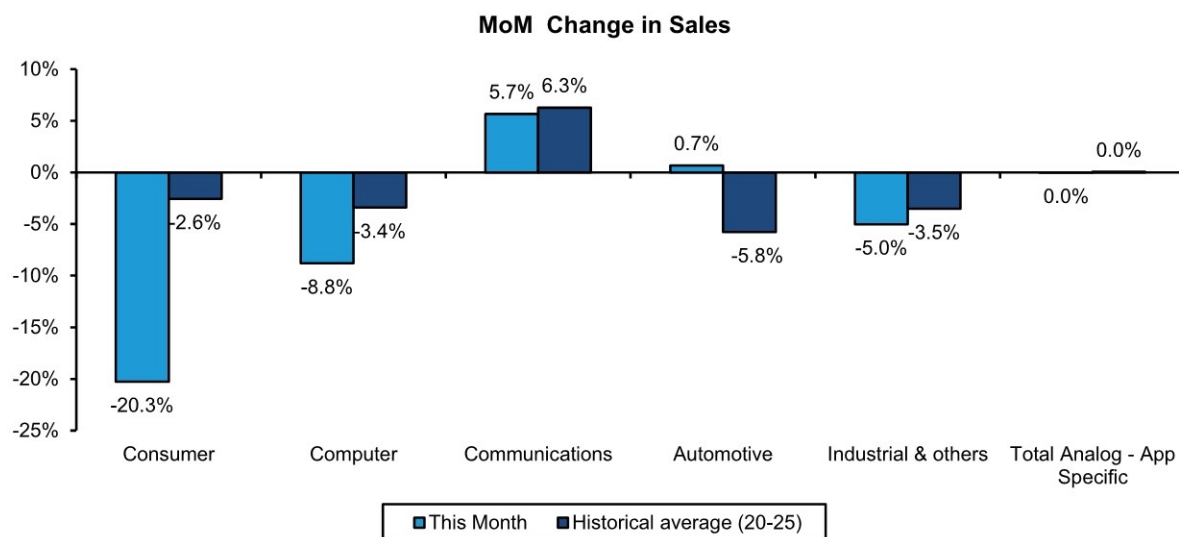
MPU: MPU sales fell -8.2% MoM, worse than typical patterns (-5.7%MoM). Units were down (-8.7%) MoM with ASPs flattish (0.5%) MoM. On a 3-months rolling basis (e.g. May-July vs. February-April), MPU sales increased (20.7%), better than typical average of 2.7%. Sales increased 18.8% vs. 3 months ago and increased 36.3% YoY.

MCU: MCU sales were down (-1.5%) MoM, better than historical averages of down -9.1%, with units down (-3.9%) and ASP up (2.4%). Consumer was down -2.5% MoM vs. typical -15.7% MoM, Computer was 23.2% MoM vs. typical -10.0% MoM, Handsets were up 20.0% MoM vs. typical of up 35.4% MoM, Wireless Communication-All Other was down -15.9% MoM vs. typical of up 0.4% MoM, Wired Communication was down -32.9% MoM vs. typical of -11.3% MoM, Automotive was up 4.3% MoM vs. typical -6.6% MoM, and Multi-purpose and others were -6.5% MoM vs. typical of -10.3% MoM (**Exhibit 15**).

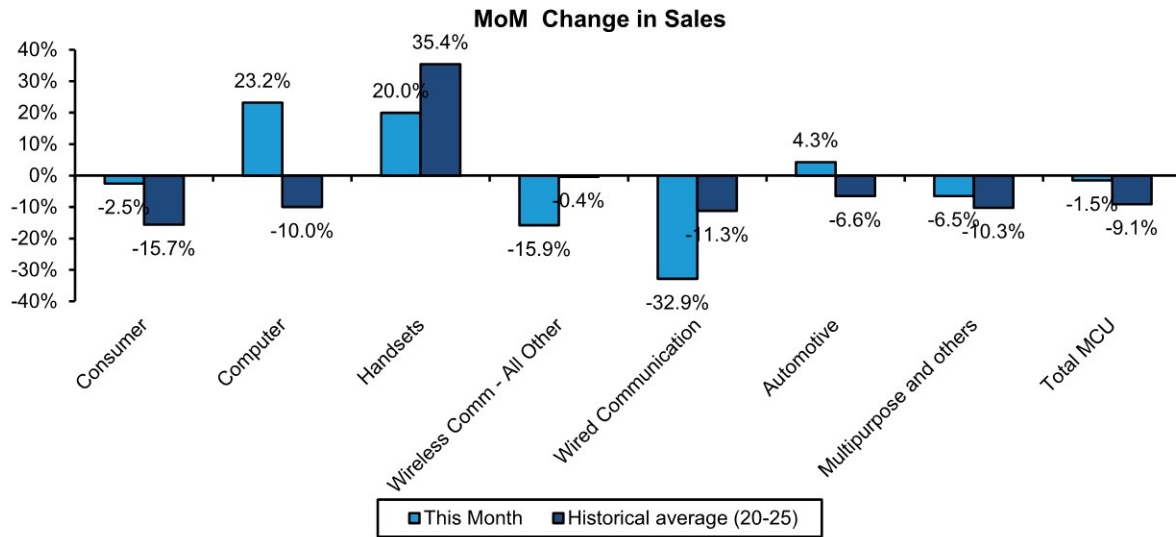
Memory: Memory sales were down (-16.3%) MoM, better than historical average of down -26.1%. On a bit shipped basis, DRAM shipments were down -22.5% MoM and NAND down -22.1% MoM. On a \$ per bit basis, DRAM ASPs rose 6.5%, while NAND ASPs rose (9.5%) (**Exhibit 16, Exhibit 17**).

EXHIBIT 13: **Analog Standard Linear - MoM change in sales**

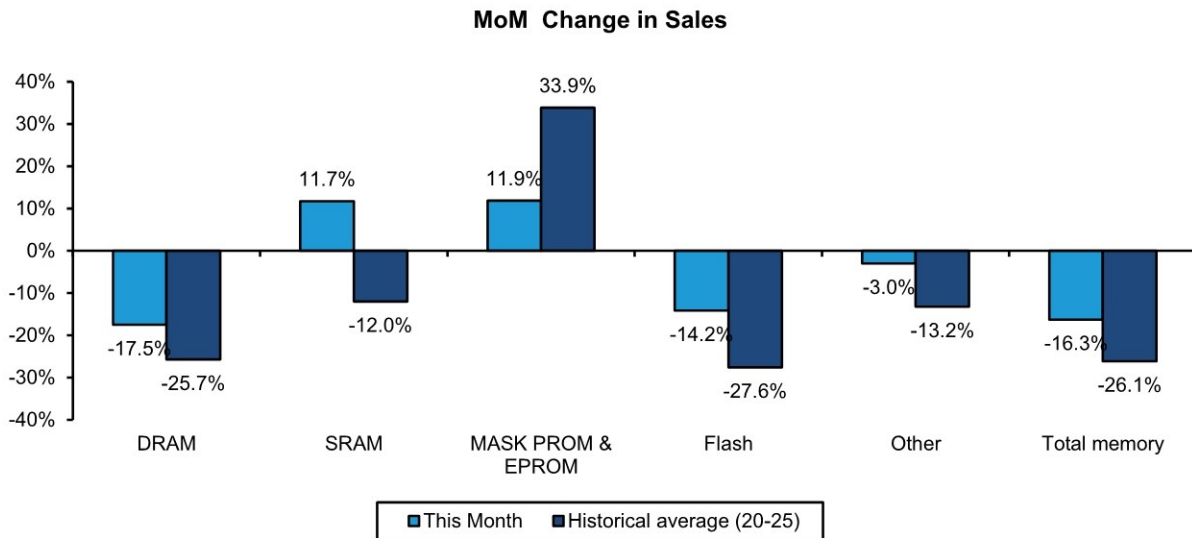
Source: WSTS, Bernstein Analysis

EXHIBIT 14: **Analog App Specific - MoM change in sales**

Source: WSTS, Bernstein Analysis

EXHIBIT 15: **MCU - MoM change in sales**

Source: WSTS, Bernstein Analysis

EXHIBIT 16: **Memory - MoM change in sales**

Source: WSTS, Bernstein Analysis

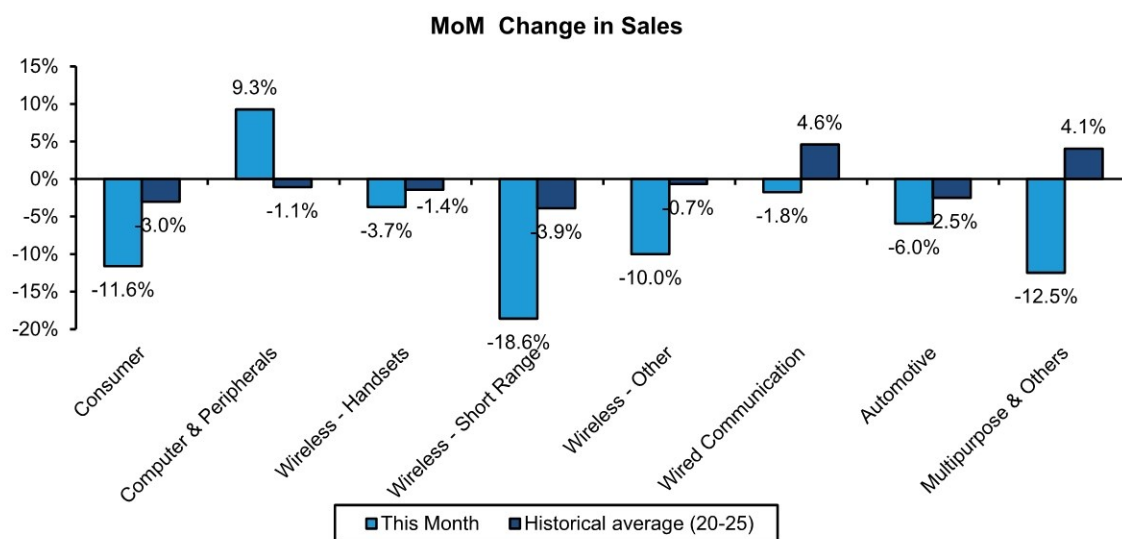
EXHIBIT 17: **Memory - Change in units and ASP on bit basis****% change based on bit shipped**

	vs 3 months			
	vs 1 month ago	months ago	rolling year ago	vs 1 year ago
DRAM	-22.5%	2.3%	6.8%	47.5%
NAND	-22.1%	3.9%	7.5%	42.3%

% change in \$ per bit

	vs 3 months			
	vs 1 month ago	months ago	ago year ago	vs 1 year ago
DRAM	6.5%	25.8%	257.9%	
NAND	9.5%	34.6%	344.1%	

Source: WSTS, Bernstein Analysis

EXHIBIT 18: **MOS Special Purpose Logic - MoM change in sales**

Source: WSTS, Bernstein Analysis

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